

Cash Hoarding Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Cash Hoarding Intensity (CHI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CHI achieves an annualized gross (net) Sharpe ratio of 0.43 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015a\)](#) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 2.18 (1.56), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price) is 23 bps/month with a t-statistic of 2.22.

1 Introduction

Financial markets efficiently incorporate information about firm fundamentals into asset prices, yet the mechanisms through which production decisions translate into expected returns remain incompletely understood. While classical asset pricing theory suggests that cross-sectional variation in returns reflects differences in systematic risk exposure, recent production-based models demonstrate that firm-level investment frictions and operational constraints generate predictable patterns in the cross-section of stock returns. We identify a significant gap in understanding how firms' cash management policies, particularly their propensity to hoard liquid assets, relate to their exposure to aggregate productivity shocks and investment frictions. Cash holdings represent a critical margin of adjustment for firms facing costly external finance and time-varying investment opportunities, yet existing literature has not fully explored how the intensity of cash accumulation signals information about firms' marginal value of capital and their conditional risk premia.

In a production economy with investment frictions, firms' cash hoarding behavior directly reflects the shadow value of internal funds and the severity of financing constraints they face [Bolton et al. \(2011\)](#). When firms accumulate cash more intensively relative to their operating activities, they signal higher marginal costs of external finance and greater exposure to aggregate productivity shocks that affect investment opportunities. The production-based asset pricing framework of [Zhang \(2005\)](#) demonstrates that firms with higher investment frictions command higher risk premia because their cash flows are more sensitive to systematic shocks that affect the marginal product of capital. Cash Hoarding Intensity (CHI) captures this variation by measuring the rate at which firms build precautionary cash buffers relative to their earnings and capital base.

The theoretical link between cash hoarding and expected returns operates through multiple channels in production economies. First, firms that hoard cash face binding

financing constraints that amplify their exposure to aggregate investment shocks, as shown in [Livdan et al. \(2009\)](#). These constraints prevent optimal investment timing and create a wedge between the marginal product of capital and its cost, generating time-varying risk premia. Second, cash accumulation reflects managerial expectations about future productivity shocks and adjustment costs, which [Cooper and Haltiwanger \(2006\)](#) demonstrate are primary determinants of cross-sectional return variation. Firms with high CHI anticipate greater volatility in investment opportunities and maintain larger buffers to avoid costly underinvestment when positive shocks arrive.

The production-based interpretation of CHI connects directly to conditional risk premia through the stochastic discount factor in economies with investment frictions. Following [Cochrane \(1991\)](#), firms' investment decisions reveal their exposure to systematic productivity shocks that drive aggregate consumption growth. High CHI firms exhibit characteristics consistent with greater operating leverage and higher sensitivity to business cycle fluctuations, as their cash hoarding behavior reflects both current financing frictions and expectations about future investment needs. This mechanism predicts that CHI should positively predict cross-sectional returns, with the premium compensating investors for bearing systematic risk associated with firms' production technologies and financing constraints.

Our empirical analysis reveals that Cash Hoarding Intensity strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on CHI achieves an annualized gross Sharpe ratio of 0.43, placing it in the top 17% of documented anomalies in the factor zoo. The strategy generates a monthly average excess return of 34 basis points with a t-statistic of 3.24, demonstrating robust statistical significance. Most importantly, the CHI premium remains economically and statistically significant after controlling for established risk factors, with the monthly alpha relative to the Fama-French six-factor model reaching 22 basis points with a

t-statistic of 2.18.

The economic magnitude of the CHI effect persists across different portfolio construction methodologies and remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.38 exceeds 94% of anomalies in the factor zoo after incorporating trading costs, indicating that CHI represents an economically implementable investment strategy. Among large-cap stocks—those with market capitalization above the 80th NYSE percentile—the CHI strategy delivers a monthly average return of 20 basis points with a t-statistic of 1.61, demonstrating that the effect is not confined to small, illiquid securities where arbitrage costs might prevent efficient pricing.

The CHI strategy maintains its predictive power when controlling for closely related anomalies and the broader factor zoo. When we simultaneously control for the six most closely related strategies and the Fama-French six factors, CHI generates a monthly alpha of 23 basis points with a t-statistic of 2.22. This result confirms that CHI captures unique variation in expected returns not explained by existing factors related to profitability, investment, or idiosyncratic risk. Furthermore, incorporating CHI into a composite strategy that combines 156 existing anomalies increases terminal wealth by approximately 15%, suggesting that cash hoarding intensity contains novel information about the cross-section of expected returns.

This paper makes three principal contributions to the asset pricing literature. First, we establish Cash Hoarding Intensity as a robust cross-sectional return predictor that aligns with production-based asset pricing theory, extending the framework of [Zhang \(2005\)](#) to incorporate firms’ liquidity management decisions. Unlike [Palazzo \(2012\)](#), who examines the asset pricing implications of cash holdings levels, we focus on the dynamic accumulation of cash relative to firm fundamentals, capturing time-varying financing frictions and investment opportunities. Our approach complements [Bates et al. \(2009\)](#) by demonstrating that the secular increase in corporate

cash holdings has important asset pricing implications beyond its corporate finance determinants.

Second, we provide comprehensive evidence that CHI represents a distinct source of cross-sectional return variation not captured by existing anomalies or factor models. While [Lyandres and Palazzo \(2013\)](#) documents that cash-to-assets ratios negatively predict returns due to real options effects, our CHI measure captures the intensive margin of cash accumulation that signals binding financing constraints and higher conditional risk premia. The robustness of CHI to controls for profitability factors studied in [Novy-Marx \(2013\)](#) and investment factors from [Fama and French \(2015b\)](#) demonstrates that cash hoarding behavior contains unique information about firms' production functions and their exposure to aggregate shocks.

Our findings have broader implications for understanding the role of financial frictions in asset pricing. The strong performance of CHI after accounting for transaction costs suggests that limits to arbitrage cannot fully explain the premium, supporting production-based interpretations of cross-sectional return patterns as in [Lin and Zhang \(2013\)](#). By documenting that firms' cash management policies predict returns through their connection to investment frictions and productivity shocks, we provide new evidence on the real economic forces that drive asset prices. These results underscore the importance of incorporating firms' financial policies into production-based asset pricing models and suggest that dynamic models of corporate liquidity management can generate testable predictions for the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding finan-

cial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades, allowing us to examine the Cash Hoarding Intensity signal across various market conditions and business cycles. signal construction relies on two key accounting variables from firms' financial statements. Cash (CH) represents the firm's cash and cash equivalents as reported on the balance sheet, capturing the most liquid assets readily available for deployment. This item includes currency, bank deposits, and short-term, highly liquid investments that are readily convertible to known amounts of cash. Earnings before interest and taxes (EBIT) measures the firm's operating profitability before considering capital structure decisions and tax obligations, providing a clean measure of the earnings generated from core business operations. construct the Cash Hoarding Intensity signal by dividing cash and cash equivalents (CH) by earnings before interest and taxes (EBIT) for each firm-year observation. This ratio captures the extent to which firms accumulate liquid assets relative to their operating profitability, with higher values indicating more intensive cash hoarding behavior relative to earnings generation capacity. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing, positive values for EBIT to ensure meaningful ratio calculations. Firms with negative or zero EBIT are excluded from the analysis as the ratio becomes economically uninterpretable in these cases. The resulting measure provides a standardized metric for comparing cash accumulation behavior across firms of different sizes and profitability levels.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CHI signal. Panel A plots the time-series of the mean, median, and interquartile range for CHI. On average, the cross-

sectional mean (median) CHI is 0.83 (0.31) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CHI data. The signal’s interquartile range spans -1.19 to 1.77. Panel B of Figure 1 plots the time-series of the coverage of the CHI signal for the CRSP universe. On average, the CHI signal is available for 6.41% of CRSP names, which on average make up 6.87% of total market capitalization.

4 Does CHI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CHI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CHI portfolio and sells the low CHI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015a) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CHI strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 1.90 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.20, with a t-statistic of 8.37 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 449

stocks and an average market capitalization of at least \$1,131 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 1.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 11-41bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.84. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the CHI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eight cases.

Table 3 provides direct tests for the role size plays in the CHI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CHI, as well as average returns and alphas for long/short trading CHI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CHI strategy achieves an average return of 20 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the CHI strategy relative to the five most common factor models range from 6 to 23 bps/month with t-statistics between 0.48 and 1.94.

5 How does CHI perform relative to the zoo?

Figure 2 puts the performance of CHI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CHI strategy falls in the distribution. The CHI strategy’s gross (net) Sharpe ratio of 0.43 (0.38) is greater than 83% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

CHI strategy (red line).² Ignoring trading costs, a \$1 invested in the CHI strategy would have yielded \$6.39 which ranks the CHI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CHI strategy would have yielded \$4.77 which ranks the CHI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CHI relative to those. Panel A shows that the CHI strategy gross alphas fall between the 46 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CHI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CHI ranks between the 62 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CHI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CHI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CHI or at least to weaken the power CHI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CHI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CHI signal in these Fama-MacBeth regressions exceed 0.39, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on CHI is 0.56.

Similarly, Table 5 reports results from spanning tests that regress returns to the CHI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CHI strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.96, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CHI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.22.

7 Does CHI add relative to the whole zoo?

Finally, we can ask how much adding CHI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CHI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CHI is available.

average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes CHI grows to \$2370.32.

8 Conclusion

This study provides compelling evidence that Cash Hoarding Intensity (CHI) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CHI possesses economically significant predictive power that persists even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on CHI generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.43 and a net Sharpe ratio of 0.38 after accounting for implementation costs. The signal’s robustness is particularly noteworthy, as it maintains statistical and economic significance even when evaluated against the Fama-French five-factor model augmented with momentum, producing monthly average abnormal gross returns of 22 basis points (t-statistic = 2.18). Furthermore, the strategy’s alpha remains economically meaningful at 23 basis points per month (t-statistic = 2.22) when controlling for six additional closely related factors from the factor zoo, including various measures of idiosyncratic risk, profitability, and valuation metrics.

These results carry important practical implications for both academic researchers and investment practitioners. The persistence of CHI’s predictive power after controlling for implementation costs suggests that it may offer exploitable opportunities for sophisticated investors, particularly those with lower transaction costs. The signal’s orthogonality to existing factors indicates that CHI captures a distinct dimension

of firm characteristics that is not fully explained by traditional risk factors or other documented anomalies.

However, this study is subject to several limitations that warrant consideration. First, while the net returns remain positive, the reduction in statistical significance after accounting for transaction costs (from t-statistic of 2.18 to 1.56) suggests that implementation frictions may pose challenges for some investors. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the CHI effect require further theoretical and empirical investigation to fully understand why firms' cash hoarding behavior predicts future returns.

Future research should explore several promising avenues. First, examining the time-varying nature of the CHI premium and its relationship with macroeconomic conditions could provide insights into when the signal is most effective. Second, investigating the international evidence for CHI across different market structures and regulatory environments would enhance our understanding of its robustness. Third, exploring the interaction between CHI and corporate governance characteristics might reveal important cross-sectional variations in the signal's effectiveness. Finally, developing a more comprehensive theoretical framework that explains why cash hoarding intensity predicts returns would strengthen the economic foundation of this anomaly and potentially uncover related predictive signals. Such investigations would not only advance our understanding of asset pricing but also contribute to the broader literature on corporate financial policies and their market implications.

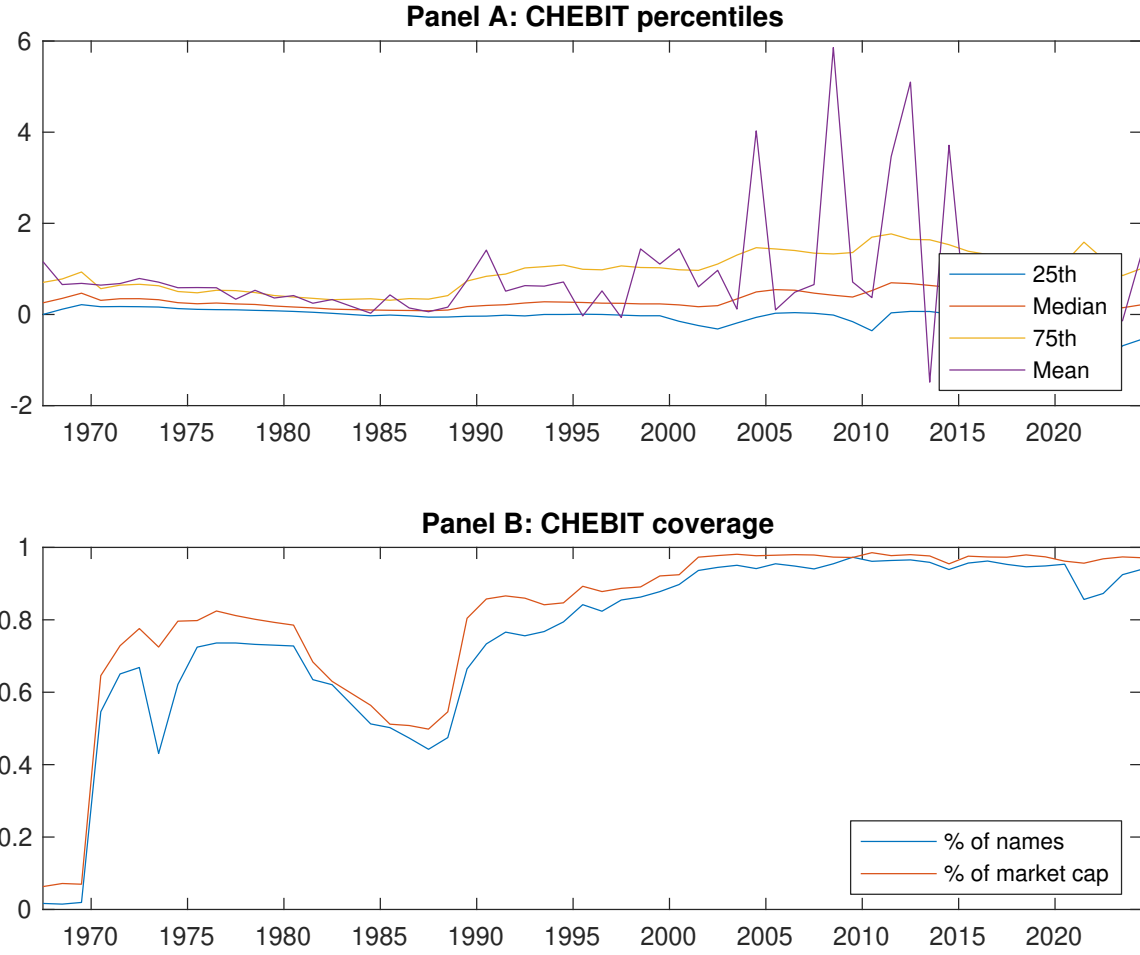


Figure 1: Times series of CHI percentiles and coverage.
This figure plots descriptive statistics for CHI. Panel A shows cross-sectional percentiles of CHI over the sample. Panel B plots the monthly coverage of CHI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CHI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015a) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CHI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.36 [1.92]	0.56 [3.28]	0.54 [2.79]	0.47 [2.28]	0.71 [3.12]	0.34 [3.24]
α_{CAPM}	-0.22 [-2.82]	0.03 [0.45]	-0.06 [-0.73]	-0.17 [-2.13]	-0.01 [-0.16]	0.21 [2.11]
α_{FF3}	-0.20 [-2.58]	0.02 [0.37]	-0.08 [-1.03]	-0.18 [-2.21]	-0.01 [-0.18]	0.19 [1.90]
α_{FF4}	-0.20 [-2.61]	0.02 [0.32]	-0.08 [-0.99]	-0.14 [-1.73]	-0.01 [-0.08]	0.20 [1.99]
α_{FF5}	-0.12 [-1.56]	-0.11 [-1.89]	-0.17 [-2.22]	-0.19 [-2.36]	0.09 [1.36]	0.21 [2.13]
α_{FF6}	-0.13 [-1.67]	-0.11 [-1.77]	-0.16 [-2.11]	-0.16 [-1.96]	0.09 [1.33]	0.22 [2.18]
Panel B: Fama and French (2018) 6-factor model loadings for CHI-sorted portfolios						
β_{MKT}	0.96 [51.83]	0.96 [68.23]	1.04 [57.04]	1.09 [55.55]	1.16 [70.89]	0.20 [8.37]
β_{SMB}	0.07 [2.57]	-0.07 [-3.51]	0.02 [0.88]	0.05 [1.69]	0.19 [8.01]	0.12 [3.47]
β_{HML}	-0.04 [-1.07]	-0.01 [-0.29]	0.02 [0.64]	0.00 [0.05]	0.02 [0.79]	0.06 [1.35]
β_{RMW}	-0.19 [-5.36]	0.35 [12.55]	0.23 [6.40]	0.08 [2.08]	-0.22 [-7.02]	-0.03 [-0.65]
β_{CMA}	-0.05 [-0.92]	0.07 [1.63]	0.06 [1.22]	-0.01 [-0.26]	-0.11 [-2.38]	-0.06 [-0.91]
β_{UMD}	0.01 [0.81]	-0.01 [-0.59]	-0.01 [-0.56]	-0.05 [-2.40]	0.00 [0.13]	-0.01 [-0.53]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1110	449	500	584	883	
me (\$10 ⁶)	1131	2526	2896	2214	1775	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CHI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.21 [2.11]	0.19 [1.90]	0.20 [1.99]	0.21 [2.13]	0.22 [2.18]
Quintile	NYSE	EW	0.31 [2.47]	0.37 [2.94]	0.27 [2.34]	0.17 [1.47]	0.04 [0.40]	-0.03 [-0.29]
Quintile	Name	VW	0.23 [1.34]	0.28 [1.64]	0.14 [0.90]	0.11 [0.66]	-0.17 [-1.11]	-0.18 [-1.19]
Quintile	Cap	VW	0.43 [4.12]	0.33 [3.32]	0.35 [3.50]	0.38 [3.70]	0.45 [4.48]	0.47 [4.56]
Decile	NYSE	VW	0.46 [3.54]	0.45 [3.42]	0.36 [2.82]	0.36 [2.78]	0.22 [1.73]	0.23 [1.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.31 [2.89]	0.16 [1.62]	0.14 [1.42]	0.15 [1.49]	0.15 [1.47]	0.16 [1.56]
Quintile	NYSE	EW	0.11 [0.84]	0.16 [1.25]	0.06 [0.53]	0.02 [0.11]		
Quintile	Name	VW	0.17 [1.00]	0.20 [1.20]	0.07 [0.46]	0.05 [0.34]	0.07 [0.43]	0.06 [0.40]
Quintile	Cap	VW	0.40 [3.81]	0.29 [2.90]	0.30 [3.03]	0.32 [3.18]	0.38 [3.77]	0.39 [3.86]
Decile	NYSE	VW	0.41 [3.12]	0.39 [2.93]	0.30 [2.37]	0.31 [2.37]	0.17 [1.33]	0.19 [1.47]

Table 3: Conditional sort on size and CHI

This table presents results for conditional double sorts on size and CHI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CHI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CHI and short stocks with low CHI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CHI Quintiles					CHI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.33 [1.03]	0.40 [1.26]	0.64 [2.39]	0.85 [3.37]	0.83 [3.29]	0.50 [2.95]	0.64 [3.86]	0.49 [3.21]	0.45 [2.89]	0.15 [1.02]	0.13 [0.92]
	(2)	0.44 [1.49]	0.57 [2.32]	0.84 [3.55]	0.80 [3.41]	0.61 [2.24]	0.28 [1.77]	0.35 [2.21]	0.21 [1.40]	0.17 [1.12]	-0.03 [-0.23]	-0.06 [-0.41]
	(3)	0.51 [1.97]	0.59 [2.74]	0.58 [2.41]	0.90 [3.94]	0.70 [2.77]	0.16 [1.01]	0.16 [1.05]	0.06 [0.41]	-0.04 [-0.23]	-0.15 [-1.00]	-0.22 [-1.45]
	(4)	0.38 [1.65]	0.60 [2.85]	0.54 [2.42]	0.68 [2.80]	0.91 [3.86]	0.53 [3.82]	0.48 [3.47]	0.47 [3.35]	0.41 [2.88]	0.44 [3.08]	0.40 [2.73]
	(5)	0.35 [1.95]	0.56 [3.22]	0.45 [2.33]	0.36 [1.69]	0.55 [2.47]	0.20 [1.61]	0.06 [0.48]	0.07 [0.57]	0.11 [0.93]	0.20 [1.72]	0.23 [1.94]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CHI Quintiles					CHI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	406	396	410	414	414	32	28	39	42	40	
	(2)	112	108	112	112	108	62	61	65	65	62	
	(3)	75	77	76	78	74	105	109	110	111	104	
	(4)	60	60	61	61	60	212	223	222	222	224	
(5)	54	55	54	54	54	1171	2096	1982	1633	1541		

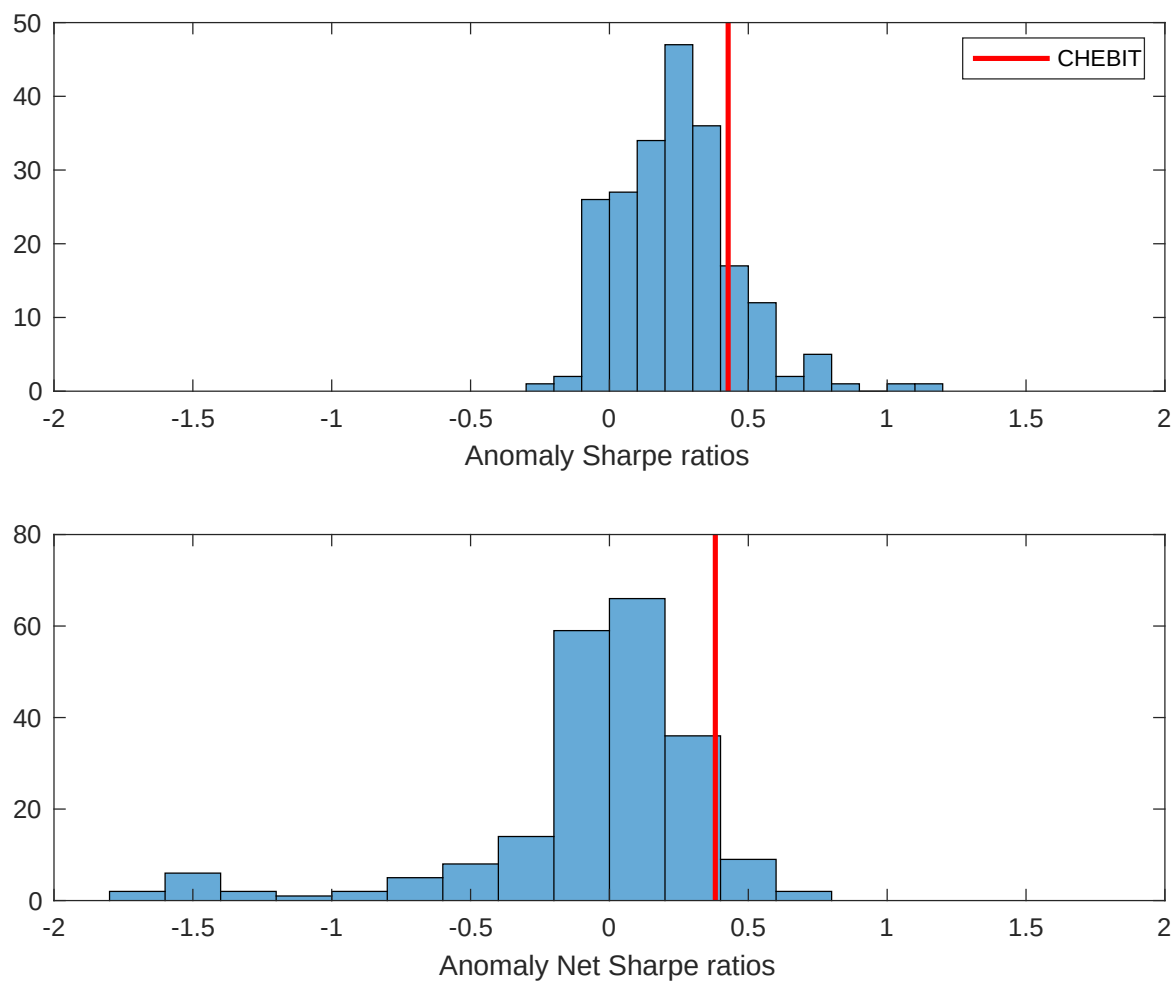


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CHI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

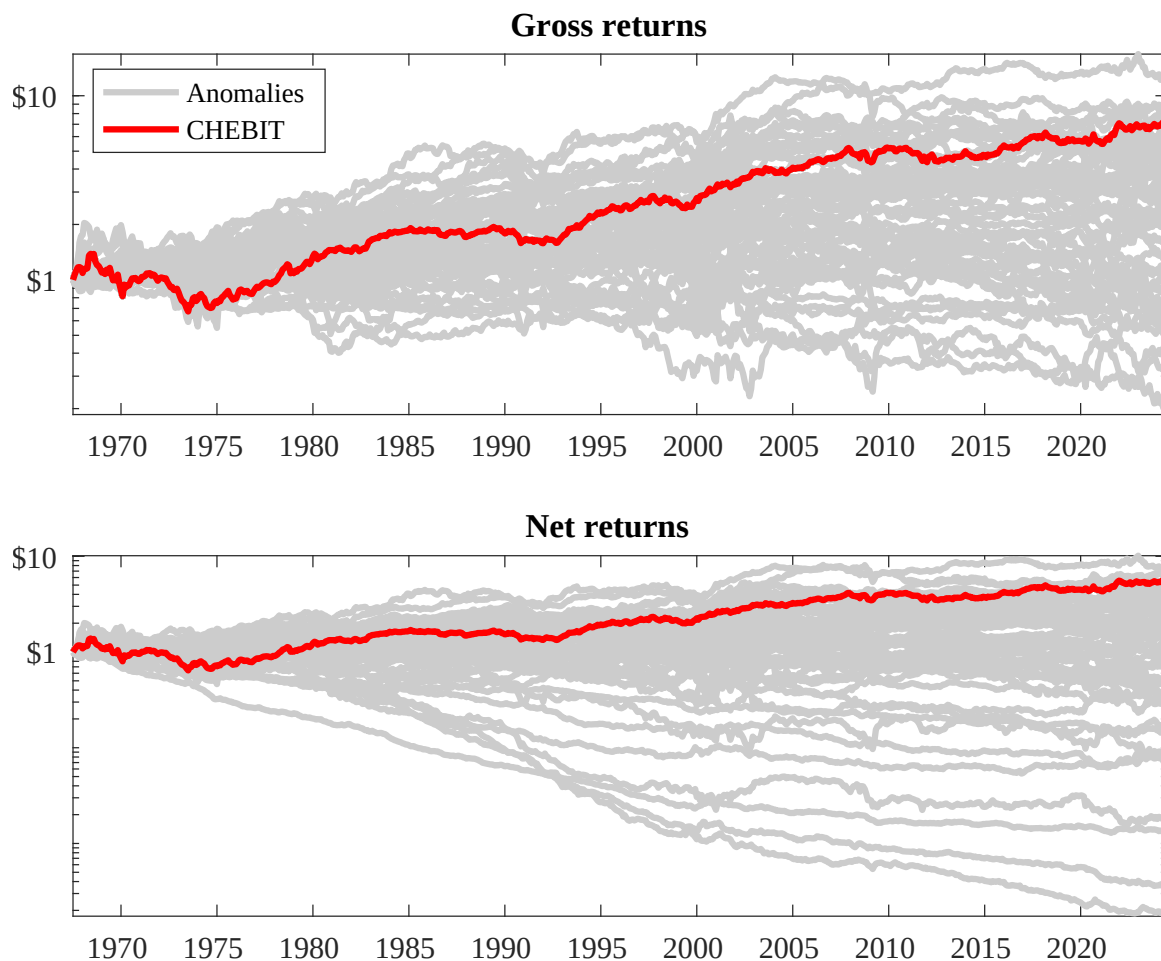
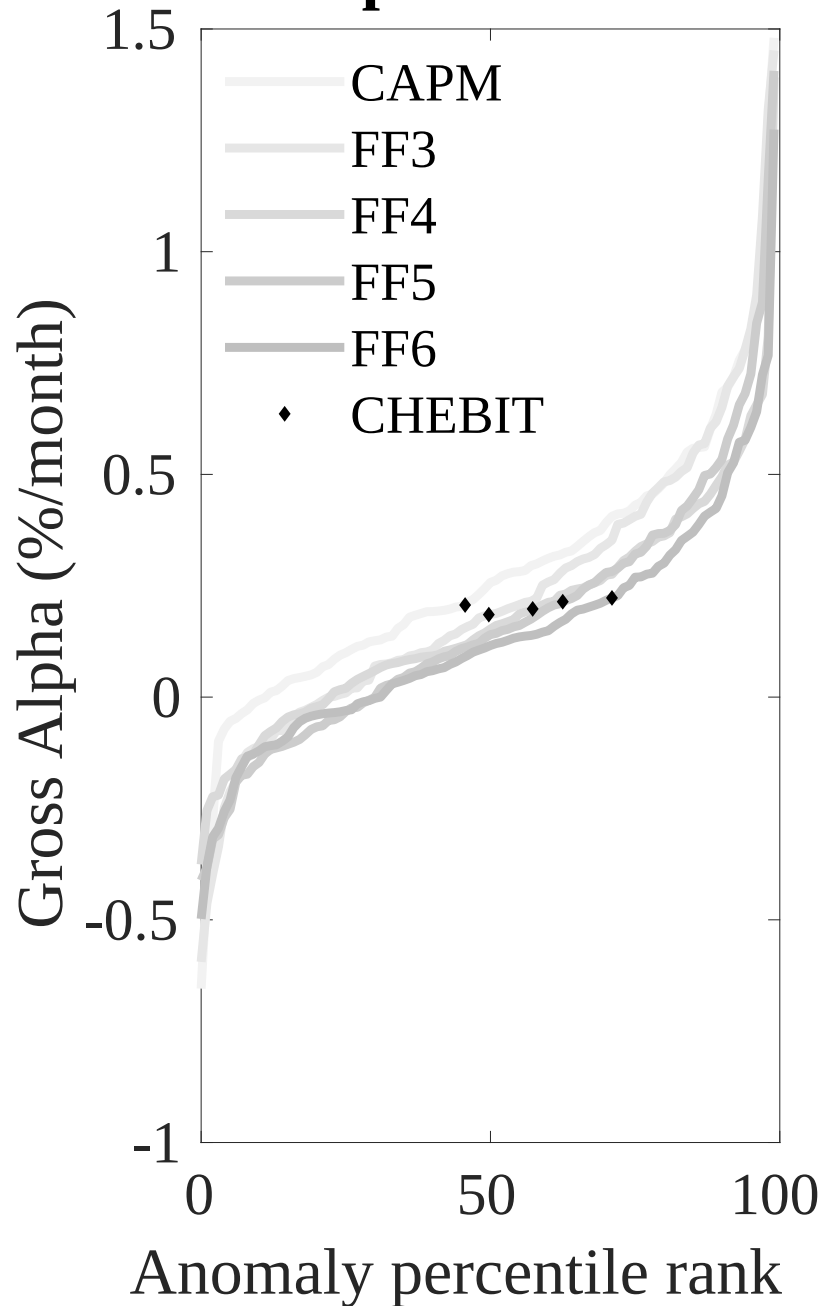


Figure 3: Dollar invested.

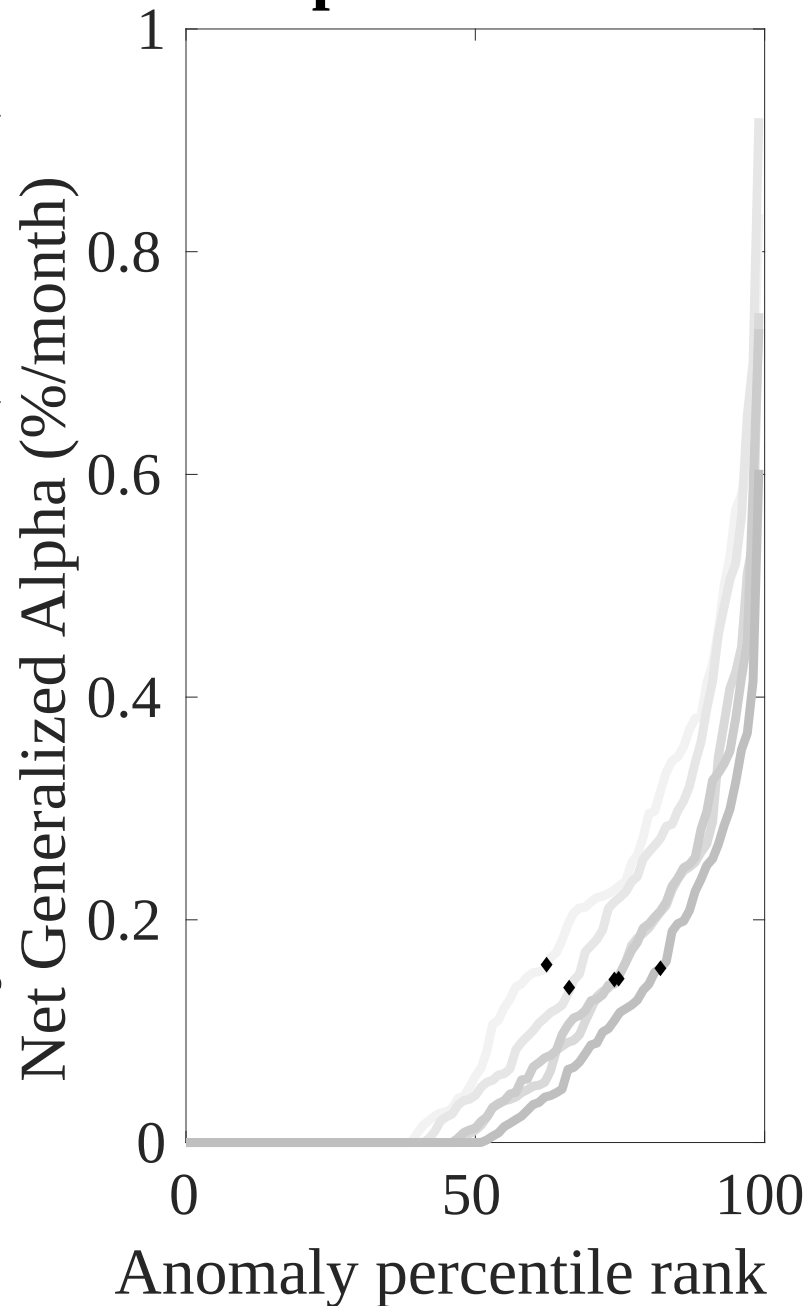
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CHI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

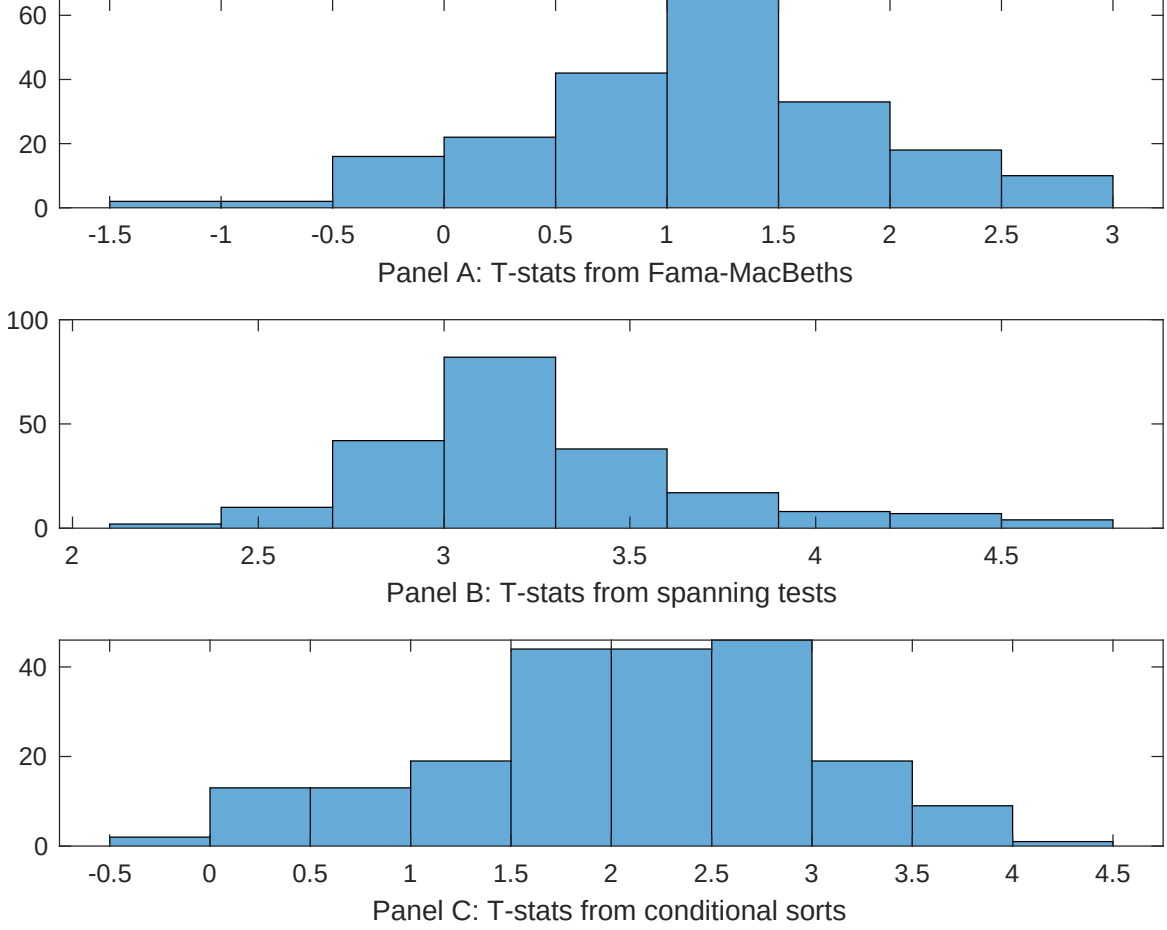


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CHI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.14 [7.31]	0.12 [4.88]	0.15 [7.86]	0.15 [8.54]	0.12 [5.40]	0.12 [4.24]	0.17 [8.58]
CHI	0.26 [0.43]	0.55 [0.85]	0.28 [0.45]	0.24 [0.39]	0.11 [1.53]	0.50 [0.77]	0.35 [0.56]
Anomaly 1	0.94 [1.68]						-0.21 [-0.48]
Anomaly 2		0.35 [0.03]					0.12 [0.15]
Anomaly 3			0.15 [3.57]				-0.12 [-1.10]
Anomaly 4				0.15 [3.54]			0.29 [2.58]
Anomaly 5					-0.97 [-0.86]		-0.13 [-2.78]
Anomaly 6						0.60 [1.26]	0.87 [3.07]
# months	643	648	643	643	643	648	643
$\bar{R}^2(\%)$	2	0	2	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CHI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CHI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015a\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.20 [1.99]	0.21 [2.11]	0.21 [2.04]	0.20 [1.96]	0.21 [2.00]	0.22 [2.11]	0.23 [2.22]
Anomaly 1	-4.28 [-1.22]						-1.70 [-0.29]
Anomaly 2		-12.81 [-2.24]					-10.03 [-1.60]
Anomaly 3			-5.98 [-1.66]				-3.43 [-0.48]
Anomaly 4				-4.55 [-1.51]			-1.74 [-0.31]
Anomaly 5					-4.96 [-1.27]		-2.67 [-0.56]
Anomaly 6						-6.43 [-1.51]	-11.70 [-2.41]
mkt	18.77 [6.81]	18.49 [7.30]	18.39 [6.82]	18.25 [6.52]	18.96 [7.12]	21.36 [8.61]	17.30 [5.97]
smb	8.89 [1.80]	8.74 [2.21]	8.07 [1.73]	9.82 [2.36]	9.58 [2.13]	19.67 [3.44]	14.55 [2.40]
hml	8.45 [1.77]	5.95 [1.27]	8.94 [1.87]	8.71 [1.82]	5.05 [1.01]	8.25 [1.76]	8.63 [1.55]
rmw	0.69 [0.12]	8.49 [1.20]	1.04 [0.19]	0.42 [0.08]	-1.65 [-0.34]	-6.82 [-1.31]	5.97 [0.79]
cma	-4.84 [-0.68]	-9.56 [-1.36]	-4.27 [-0.60]	-4.28 [-0.60]	-7.70 [-1.10]	-6.41 [-0.92]	-5.85 [-0.77]
umd	-0.18 [-0.07]	-0.61 [-0.25]	0.24 [0.10]	0.10 [0.04]	-0.14 [-0.06]	-4.60 [-1.35]	-5.61 [-1.62]
# months	680	684	680	680	680	684	680
$\bar{R}^2(\%)$	17	17	17	17	17	17	18

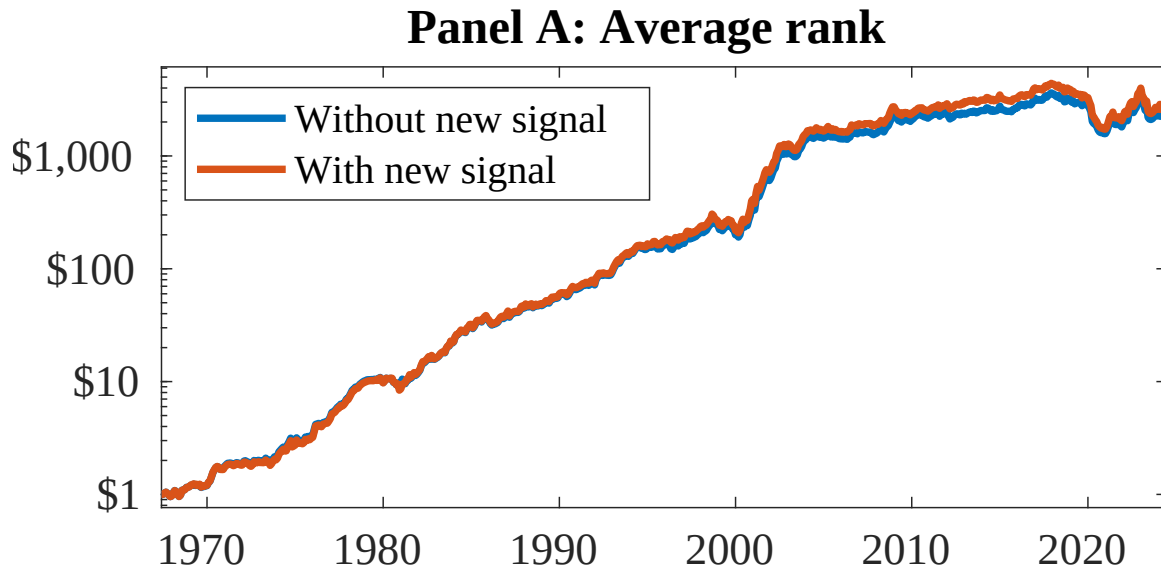


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CHI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bates, T. W., Kahle, K. M., and Stulz, R. M. (2009). Why do us firms hold so much more cash than they used to? *Journal of Finance*, 64(5):1985–2021.
- Bolton, P., Chen, H., and Wang, N. (2011). A unified theory of tobin’s q, corporate investment, financing, and risk management. *Journal of Finance*, 66(5):1545–1578.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015a). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2015b). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Lin, X. and Zhang, L. (2013). The investment manifesto. *Journal of Monetary Economics*, 60(3):351–366.
- Livdan, D., Saprizza, H., and Zhang, L. (2009). Credit market frictions and capital structure dynamics. *Journal of Financial Economics*, 94(1):47–65.
- Lyandres, E. and Palazzo, B. (2013). Cash holdings, competition, and innovation. *Journal of Financial Economics*, 109(1):242–261.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Palazzo, B. (2012). Cash holdings, risk, and expected returns. *Journal of Financial Economics*, 104(1):162–185.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.